6.2 DNA replication

Genetic information has to flow from the parental cells or organisms to progeny by a very faithful mechanism. This copy of DNA from DNA known as replication is the first and fundamental step in the central dogma. Any DNA replication mechanisms should consider and overcome the following complexities

1. As we have studied already the strands of DNA run in opposite directions (5'->3') on both strands. So the new nucleotides added should maintain that anti parallel directionality.
2. DNA is helical and a supply of energy is required to unwind it.
3. Such unwound single strands should be prevented from base pairing once again (re-annealing)
4. The DNA molecule is large and so the replication process should start and proceed at several points.
5. There should be a special mechanism to identify and correct errors during replication.

The replication of DNA molecule is an enzymatic reaction catalyzed by the enzyme DNA polymerase. However several other protein molecules are also associated with the replication process considering the above mentioned complexities.

6.2.1 The models of DNA replication

After the discovery of Watson and Crick double helical DNA structure, three different types of models for DNA replication were proposed, which were the conservative, semi conservative and dispersal models.

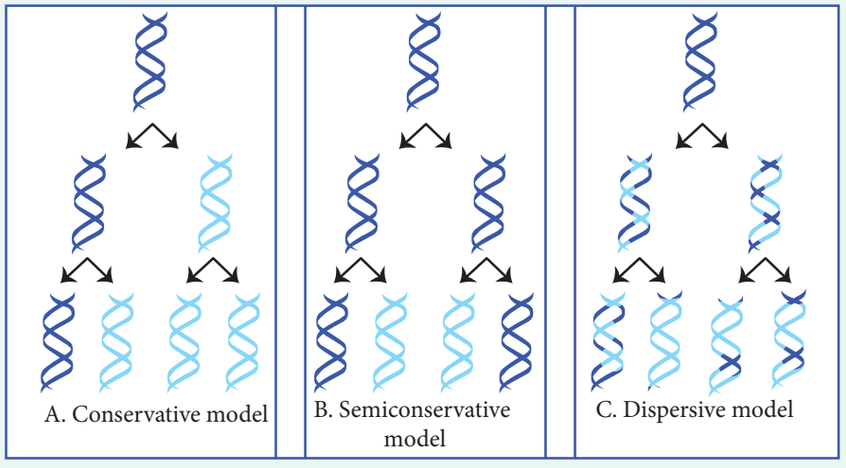


Figure 2 Different models of DNA replication

6.2.2 The conservative model

In this model the two parental strands base pair once again after forming a new copy. The newly synthesized strands combine together to form another daughter molecule. So in effect at the end of one cycle of replication an intact parental DNA and a new daughter DNA are obtained.

6.2.3 The Semiconservative model

According to this model the two strands separate in parental DNA and each of these strands act as a template to form a new DNA molecule. Thereby, after a cycle of replication

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